

Database Management Systems: Assignment-1

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Q1: Create a table EMPLOYEE with following schema:

(Emp_no, E_name, E_address, E_ph_no, Dept_no, Dept_name, Job_id , Salary)

1. Add a new column; HIREDATE to the existing relation.

Query:

```
ALTER TABLE employee  
ADD HIREDATE date NOT NULL;
```

Output:

Field	Type	Null	Key	Default	Extra
Emp_no	char(10)	NO	PRI	NULL	
E_address	varchar(50)	NO		NULL	
E_ph_no	varchar(10)	NO		NULL	
Dept_no	char(10)	NO		NULL	
Dept_name	varchar(10)	NO		NULL	
Job_id	char(10)	NO		NULL	
Salary	decimal(8,2)	NO		NULL	
HIREDATE	date	NO		NULL	

2. Change the datatype of JOB_ID from char to varchar.

Query:

```
ALTER TABL employee  
MODIFY job_id varchar(10) NOT NULL;
```

Output:

Field	Type	Null	Key	Default	Extra
Emp_no	char(10)	NO	PRI	NULL	
E_address	varchar(50)	NO		NULL	
E_ph_no	varchar(10)	NO		NULL	
Dept_no	char(10)	NO		NULL	
Dept_name	varchar(10)	NO		NULL	
job_id	varchar(10)	NO		NULL	
Salary	decimal(8,2)	NO		NULL	
HIREDATE	date	NO		NULL	

3. Change the name of column/field Emp_no to E_no.

Query:

```
ALTER TABLE employee  
CHANGE COLUMN Emp_no E_no char(10);
```

Output:

Field	Type	Null	Key	Default	Extra
E_no	char(10)	NO	PRI	NULL	
E_address	varchar(50)	NO		NULL	
E_ph_no	varchar(10)	NO		NULL	
Dept_no	char(10)	NO		NULL	
Dept_name	varchar(10)	NO		NULL	
job_id	varchar(10)	NO		NULL	
Salary	decimal(8,2)	NO		NULL	
HIREDATE	date	NO		NULL	

4. Modify the column width of the job field of emp table

Query:

```
ALTER TABLE employee  
MODIFY job_id varchar(20) NOT NULL;
```

Output:

Field	Type	Null	Key	Default	Extra
E_no	char(10)	NO	PRI	NULL	
E_address	varchar(50)	NO		NULL	
E_ph_no	varchar(10)	NO		NULL	
Dept_no	char(10)	NO		NULL	
Dept_name	varchar(10)	NO		NULL	
job_id	varchar(20)	NO		NULL	
Salary	decimal(8,2)	NO		NULL	
HIREDATE	date	NO		NULL	

Q2: Create following tables as per given schema:

1. Create Table "employee"

create table (empid varchar(10) primary key, empname varchar(20) not null,
gender varchar(7) not null, age number(3) not null, dept varchar(15) not null,
dob date not null, doj date not null);

2. Create table "salary"

create table salary (empid varchar(10) references employee(empid), salary
number(10) not null, dept varchar(15) not null, branch varchar(20) not null);

3. Create table "branchtable"

create table branchtable (branch varchar(20) not null, city varchar(20) not null);

Perform the following operations:

1. Add a new column named designation in the employee table.

Query:

```
ALTER TABLE employee  
ADD designation varchar(20) NOT NULL;
```

Output:

Field	Type	Null	Key	Default	Extra
empid	varchar(10)	NO	PRI	NULL	
empname	varchar(20)	NO		NULL	
gender	varchar(7)	NO		NULL	
age	int(3)	NO		NULL	
dept	varchar(15)	NO		NULL	
dob	date	NO		NULL	
doj	date	NO		NULL	
designation	varchar(20)	NO		NULL	

2. To drop a column from table salary.

Query:

```
ALTER TABLE salary  
DROP COLUMN branch;
```

Output:

Field	Type	Null	Key	Default	Extra
empid	varchar(10)	YES	MUL	NULL	
salary	decimal(10,0)	NO		NULL	
dept	varchar(15)	NO		NULL	

3. Rename the table employee to emp.

Query:

```
ALTER TABLE employee  
RENAME TO emp;
```

Output:

```
MariaDB [assignment_1]> desc emp;
```

Field	Type	Null	Key	Default	Extra
empid	varchar(10)	NO	PRI	NULL	
empname	varchar(20)	NO		NULL	
gender	varchar(7)	NO		NULL	
age	int(3)	NO		NULL	
dept	varchar(15)	NO		NULL	
dob	date	NO		NULL	
doj	date	NO		NULL	
designation	varchar(20)	NO		NULL	

4. Display the tables.

Query:

```
DESC emp;
```

Output:

```
MariaDB [assignment_1]> desc emp;
```

Field	Type	Null	Key	Default	Extra
empid	varchar(10)	NO	PRI	NULL	
empname	varchar(20)	NO		NULL	
gender	varchar(7)	NO		NULL	
age	int(3)	NO		NULL	
dept	varchar(15)	NO		NULL	
dob	date	NO		NULL	
doj	date	NO		NULL	
designation	varchar(20)	NO		NULL	

Query:

DESC salary;

Output:

```
MariaDB [assignment_1]> desc salary;
```

Field	Type	Null	Key	Default	Extra
empid	varchar(10)	YES	MUL	NULL	
salary	decimal(10,0)	NO		NULL	
dept	varchar(15)	NO		NULL	

Query:

DESC branchtable;

Output:

```
MariaDB [assignment_1]> desc branchtable;
```

Field	Type	Null	Key	Default	Extra
branch	varchar(20)	NO		NULL	
city	varchar(20)	NO		NULL	

5. Perform drop and truncate operations.

Query:

DROP TABLE branchtable;

Output:

```
MariaDB [assignment_1]> desc branchtable;  
ERROR 1146 (42S02): Table 'assignment_1.branchtable' doesn't exist
```

Input:

```
MariaDB [assignment_1]> select * from salary;
```

empid	salary	dept
E101	50000	HR
E102	60000	IT
E103	55000	Finance

Query:

TRUNCATE TABLE salary;

Output:

```
MariaDB [assignment_1]> select * from salary;  
Empty set (0.001 sec)
```